

Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

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Abstract

We study expensive chance-constrained simulation optimization for ordered policy profiles with historical source tasks. Before any target outcome, our method ranks a public 64-profile library from replicated source objective and chance-margin outcomes, then chooses ten profiles by farthest-first coverage in an exact cosine-profile coordinate. Target search and independent verification remain replaceable, so the contribution is an initial design, not a new Bayesian-optimization backend. We establish cross-grid coordinate consistency, a factor-two finite-library covering guarantee, source-rank recovery, conditional source-to-target coverage, and exact familywise safety for a frozen shortlist. In a registered randomized task law with eight aligned and adverse regimes, 160 independently seeded latent tasks crossed with three grid resolutions (480 task-resolution cells per design), the source-scored design finds a truly feasible policy in 91.7% of tasks and an independently certified one in 47.3%, versus 37.5% and 10.6% for an outcome-free cosine maximin design. The comparison reverses in aligned, irregular-grid, sparse-high-frequency, and misspecified regimes. In a region-held-out electricity-reserve study, source scoring certifies 60/90 runs versus 70/90 for the generic design; target-only functional optimization also wins at equal one-target preverification cost. Historical simulation is therefore justified only as an amortized, task-law-dependent design investment.

Keywords: simulation optimization; chance constraints; initial design; transfer learning; functional decisions; independent verification

1 Introduction

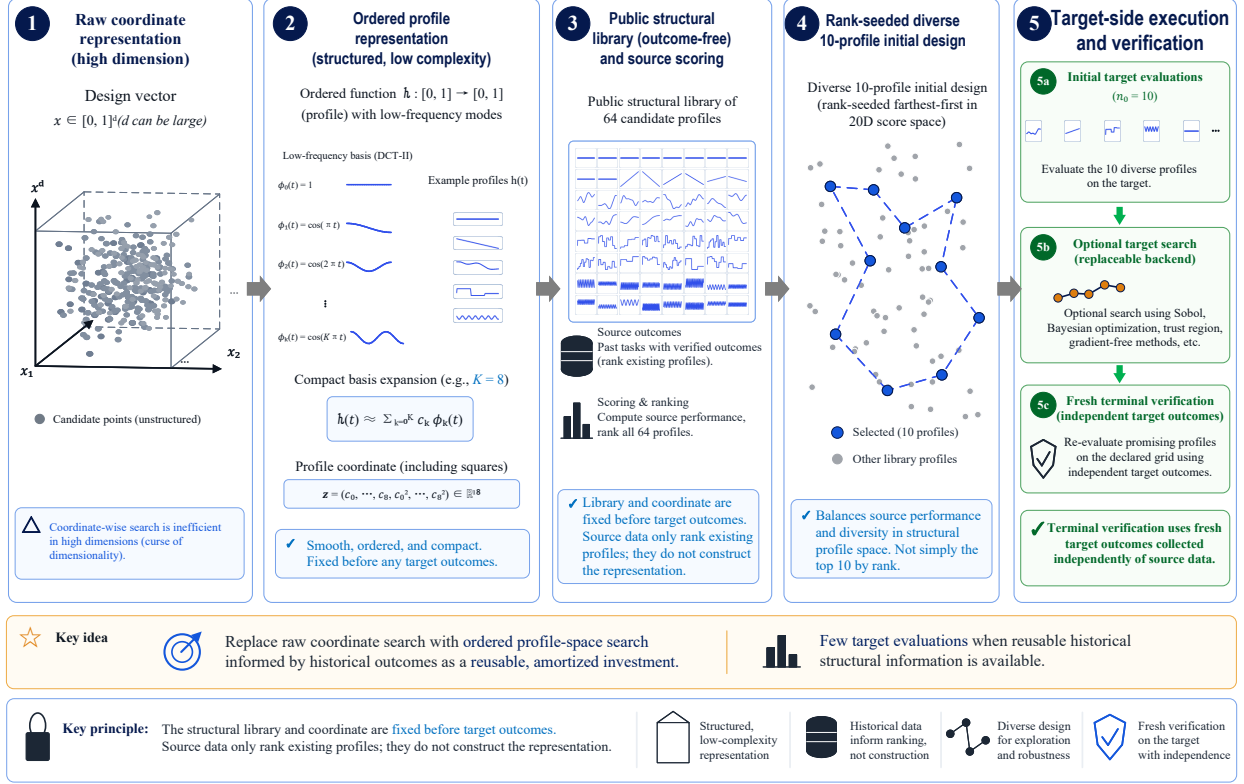
Many simulation-optimization decisions are functions before they are vectors. A reserve rule varies with forecast stress, an inventory rule varies across a planning horizon, and a

resource rule varies over ordered operating states. Implementation discretizes such a policy into hundreds or thousands of coordinates, but neighboring coordinates retain a common order and often a smooth or piecewise-smooth interpretation. Treating this representation as an arbitrary high-dimensional box discards information that can matter more than the choice of sequential optimizer.

The difficulty is sharper under chance constraints. With only a few target simulations, an unstructured design may never observe a feasible policy. A surrogate or acquisition function then has little evidence with which to learn the feasible boundary. Historical simulations from related tasks appear to offer a remedy, but they introduce two risks. First, source tasks can be misaligned: a shape that is safe in one system may be harmful in another. Second, source cost and final safety verification can dwarf the advertised target-search budget. A credible method must therefore expose its structural assumption, separate source information from target information, and permit abstention when independent evidence is insufficient.

We study a deliberately narrow question: can replicated source outcomes improve the *initial design* for a new chance-constrained ordered-profile problem? We begin with a finite, outcome-free library of constant, monotone, low-frequency, piecewise, and high-frequency profiles on $[0, 1]$. Source outcomes do not create this library. They score its members using within-task objective and chance-margin ranks. A first profile balances the two ranks; Gonzalez farthest-first selection then fills the design in a coordinate that contains exact cosine coefficients, their diagonal squares, and the two source ranks. The ten selected profiles are frozen and mapped to the target grid before any target outcome is observed. We call the result a *source-scored structural initial design*.

This construction separates three decisions that are often conflated. The frontend decides where target search starts. A replaceable backend may add target evaluations. A frozen ordered shortlist is then tested on fresh replications by an optimizer-independent verifier. Figure 1 summarizes the central change in viewpoint: the method does not confront the large discretized coordinate space solely with a more elaborate acquisition rule. It first uses the ordered functional interpretation to fix a public structural profile class, then uses historical simulations only to rank and diversify the initial design. Verification outcomes never return to search.



DCT: discrete cosine transform; 20D: 18 profile coordinates + (optionally) objective and risk scores.

Figure 1: From high-dimensional coordinates to ordered profile-space search. A discretized vector is treated as an ordered policy profile and represented by low-frequency cosine coefficients. A public, outcome-free structural library and its coordinate are fixed before target outcomes. Replicated source outcomes rank existing profiles but do not construct the representation. Rank-seeded farthest-first selection chooses a diverse ten-profile initial design. A replaceable backend may add target evaluations, after which fresh independent samples verify a frozen shortlist. Historical simulation therefore supplies reusable experimental-design information rather than target-oracle information.

Our contributions are fourfold.

- (i) We give a complete finite algorithm for turning source simulations into an initial design for ordered profiles. Every library rule, coordinate, weight, tie break, interpolation, and information boundary is explicit. The method uses no target objective, feasibility label, optimum, safe-basin parameter, or verification outcome before target search.
- (ii) We connect the implementation to a finite theoretical chain. The cosine coordinate is stable under profile-grid refinement; farthest-first is a two-approximation for finite k -center coverage; bounded source estimation error preserves sufficiently separated ranks; and profile coverage implies target feasibility under explicit source-target discrepancy and safety-depth conditions. A separate exact-binomial result controls familywise false deployment for the frozen shortlist. The finite implications are machine-checked in Lean

4; the alignment assumptions remain empirical scope conditions rather than universal claims.

- (iii) We conduct a frozen stress test designed to reveal benchmark fit. Eight randomized regimes vary effective rank, frequency support, coordinate order, grid regularity, smoothness, sparse high-frequency activity, and complete source-target misspecification. Structured source-free controls use the same finite library and profile coordinate. Additional matrices vary source budget, representation schema, chance level, safety mass, and method constants.
- (iv) We account for source, target-search, and verification calls separately. An equal preverification-cost experiment gives target-only methods the source archive’s calls. A region-held-out electricity study is retained even though it reverses the source-versus-generic ordering. Native transfer methods use their own pipelines rather than receiving our initial design. These negative and equal-cost results delimit the contribution: source scoring is useful under aligned task laws and repeated deployment, not as a universal shortcut for arbitrary high-dimensional optimization.

The main empirical result is correspondingly conditional. Across 160 independently seeded latent tasks crossed with three profile-grid resolutions (480 paired task–resolution cells per design), source scoring improves aggregate feasible coverage and certification. It loses, however, in several registered regimes and in the electricity study. A 10,000-coordinate result is therefore interpreted as refinement of an ordered profile grid, not as solving an arbitrary 10,000-variable black-box problem. This boundary is central to both the theory and the practical decision of whether historical simulation is worth buying.

2 Related Literature

Simulation optimization and noisy constraints. Simulation optimization studies expensive stochastic systems for which objectives or constraints are available only through simulation (Fu, 2002; Amaran et al., 2016). Stochastic kriging explicitly separates response uncertainty from simulation noise (Ankenman et al., 2010), while constrained Bayesian optimization models noisy objective and feasibility observations jointly (Letham et al., 2019). Our question occurs one stage earlier: before fitting a target surrogate, which few ordered profiles should receive the first simulations?

High-dimensional and functional optimization. Trust-region methods and sparse axis-aligned priors can make Bayesian optimization more robust in large ambient spaces (Eriksson et al., 2019; Eriksson and Poloczek, 2021; Eriksson and Jankowiak, 2021). They do not by

themselves encode that coordinates discretize one ordered function. Functional Bayesian optimization instead searches over function-valued decisions (Vien et al., 2018). Functional-data analysis likewise represents curves through common bases and treats discretization as observation of a function rather than unrelated coordinates (Ramsay and Silverman, 2005). Our profile coordinate is less flexible but fully finite: it maps irregular or refined grids into a common cosine summary, then selects from a declared structural library. Nominal grid dimension and effective profile complexity are therefore reported separately.

Space-filling initial designs. Maximin and minimax distance criteria are classical model-free devices for covering an experimental region (Johnson et al., 1990), and space-filling designs are standard for deterministic computer experiments (Morris and Mitchell, 1995). Gonzalez farthest-first gives a factor-two finite k -center guarantee (Gonzalez, 1985). We use that algorithm on a finite profile library, but source ranks augment its selection metric. The generic-DCT control removes those ranks and is therefore the direct space-filling counterfactual.

Transfer and historical-task priors. Few-shot deep kernels, hierarchical Gaussian-process priors, source ensembles, and meta-learned acquisitions transfer different objects across tasks (Wistuba and Grabocka, 2021; Rothfuss et al., 2021, 2023; Wang et al., 2024; Tighineanu et al., 2022; Volpp et al., 2020; Pan et al., 2024). Related work also learns a search space from earlier hyperparameter tasks (Perrone et al., 2019) or transfers optimizer rankings (Feurer et al., 2018). Warm-start Bayesian optimization similarly treats earlier evaluations as reusable information (Poloczek et al., 2016). In contrast, our source archive scores a fixed public library; it does not learn a target surrogate, acquisition function, or hidden target-specific representation, and it assumes the archive has already been retrieved. This narrower object makes source cost and the source-free structural counterfactual directly auditable. We nevertheless compare eight native end-to-end transfer pipelines using the same source information and target budget.

Selection and independent verification. Chance-constrained ranking-and-selection procedures allocate simulation to identify a feasible best system while controlling selection error (Hong et al., 2015). Post-optimization ranking and selection can also “clean up” candidate solutions generated by a heuristic search (Boesel et al., 2003). We adopt this separation literally. Search produces an ordered shortlist; fresh Bernoulli feasibility replications test its members; and Bonferroni spending composes candidate-wise exact tests. The all-success rule is conservative, so we report power and verification cost in addition to false certificates. This differs from treating a surrogate’s posterior feasibility probability as a

deployment certificate.

Positioning. Without restrictions on the objective and feasible set, no finite target-label-free design can dominate all others, a point consistent with no-free-lunch results (Wolpert and Macready, 1997). Our contribution is not a distribution-free optimizer. It is a task-law-conditional experimental-design protocol for ordered profiles, accompanied by source-free structured controls, adverse regimes, a negative external control, and explicit cost accounting.

3 Problem and Information Contract

3.1 Ordered policy profiles

Let $h : [0, 1] \rightarrow [0, 1]$ denote an ordered policy profile. A task q supplies ordered nodes

$$0 < t_{q,1} < \dots < t_{q,d_q} < 1$$

and an integer implementation scale L_q . The deployed vector is

$$x_{q,i}(h) = \text{round}\{L_q h(t_{q,i})\}, \quad i = 1, \dots, d_q.$$

Thus d_q is a grid resolution, not by itself the intrinsic dimension of the admissible profile class. Regular grids use $t_{q,i} = (i - 1/2)/d_q$; irregular grids are allowed when their ordering is declared. A schema may additionally declare channel order. No claim is made for an unordered vector with unknown coordinate semantics.

One simulator replication at x returns an objective observation $Y_q^f(x)$ and a scalar constraint observation $Y_q^g(x)$. We minimize

$$f_q(x) = \mathbb{E}[Y_q^f(x)]$$

subject to the chance constraint

$$p_q(x) := \mathbb{P}\{Y_q^g(x) \leq \tau_q\} \geq 1 - \alpha_q. \tag{1}$$

The true feasible set is $\mathcal{F}_q = \{x : p_q(x) \geq 1 - \alpha_q\}$. Truth is available in synthetic experiments only after all design, search, shortlisting, and verification decisions, and only for audit metrics.

3.2 Source and target information

Let $q_1, \dots, q_m$ be source tasks and $q_\star$ a target task. Before the target opens, every source task evaluates the same finite normalized profile library $\mathcal{L} = \{h_1, \dots, h_J\}$. For each pair (q, h) , the source archive contains R ordinary replications of (Y_q^f, Y_q^g) . The primary method uses $m = 2$, $J = 64$, and $R = 3$, hence

$$S = mJR = 384 \tag{2}$$

source simulator calls. The library is constructed without source or target outcomes; source data only determine which members are selected.

The source archive is an exogenously supplied input. In the synthetic stress test its tasks are drawn from the same registered regime as the target; in the external study they obey the stated region-holdout rule. The method ranks profiles within that supplied archive. It does not identify, retrieve, or purchase related historical tasks from a larger repository. Archive retrieval is therefore an upstream decision outside the claimed algorithm.

Before target search, the method may use the target’s ordered grid, integer bounds, nominal dimension, and, in declared-schema experiments, channel order. It may not use a target outcome, feasibility label, noise parameter, optimum, safe-basin center, or verification sample. The primary source aggregation is domain blind: source tasks receive equal weight. Descriptor-conditioned weighting is evaluated only as a control.

3.3 Decision stages and costs

The frontend selects n_0 initial target policies. A backend may then use a total of $N \geq n_0$ target search calls and returns an ordered shortlist of at most K_s distinct candidates. The shortlist and its order are frozen before verification. Fresh replications test the candidates, producing either a certified deployment or abstention.

We report the costs

$$C_{\text{all}} = S + N + V, \quad C_{\text{amort}}(M) = \frac{S}{M} + N + V, \tag{3}$$

where V is the realized independent-verification cost and M is the number of target deployments sharing one source archive. “Equal preverification cost” means that $S + N$ is matched before V ; it is not called equal total cost.

3.4 Endpoints

The primary search endpoint is whether the N target evaluations contain a truly feasible policy. The primary deployment endpoint is whether the verifier returns a truly feasible policy. We also report false certificates, verification calls, and a penalized loss that retains failures in the denominator. Conditional regret is reported only when a feasible policy exists and is never used to compare methods with different feasible counts by itself. A fixed, outcome-free audit library defines synthetic regret and an explicit oracle upper bound; neither enters a non-oracle decision.

4 Source-Scored Structural Initial Design

This section specifies the complete frontend. The implementation contains no additional scalarization, endpoint replacement, or target-dependent repair.

4.1 Outcome-free profile library

All tasks share a deterministic library $\mathcal{L} = \{h_1, \dots, h_{64}\}$ on 128 regular cell midpoints. Its seed is 20260808. It contains:

- ten constants with levels $0.05, 0.15, \dots, 0.95$;
- six linear ramps with endpoint pairs $(.05, .95), (.95, .05), (.20, .80), (.80, .20), (.35, .65)$, and $(.65, .35)$;
- sixteen low-frequency cosine profiles, with rank drawn uniformly from $\{2, \dots, 8\}$, coefficient $a_k \sim N(0, .12^2/k)$, and baseline drawn uniformly from $[.25, .75]$;
- sixteen piecewise-constant profiles with 3–12 blocks and levels drawn uniformly from $[.08, .92]$, followed by the declared deterministic smoothing;
- sixteen cosine profiles with frequency drawn uniformly from $\{9, \dots, 40\}$, baseline in $[.30, .70]$, amplitude in $[.08, .25]$, and a uniform phase.

Values are clipped to $[0, 1]$. Duplicate profiles are skipped, and stable profile identifiers follow construction order. This mixture is a structural prior. It is public, generated before any outcome, and is not claimed to span all useful policies.

4.2 Grid-invariant profile coordinate

For ordered nodes $t_1 < \dots < t_d$, let $0 = e_0 < e_1 < \dots < e_d = 1$ be their Voronoi cell edges, where $e_i = (t_i + t_{i+1})/2$ internally. For the piecewise-constant reconstruction of h , define

$$c_0(h) = \sum_{i=1}^d h(t_i)(e_i - e_{i-1}), \quad (4)$$

$$c_k(h) = \sqrt{2} \sum_{i=1}^d h(t_i) \frac{\sin(\pi k e_i) - \sin(\pi k e_{i-1})}{\pi k}, \quad k = 1, \dots, K. \quad (5)$$

These are exact integrals of the cosine basis over the reconstruction cells, not midpoint approximations to the basis. With $K = 8$ and $\kappa = .25$, the profile coordinate is

$$\phi(h) = (\tilde{c}_0, \dots, \tilde{c}_K, \tilde{c}_0^2, \dots, \tilde{c}_K^2), \quad \tilde{c}_k = \frac{c_k}{1 + \kappa k}. \quad (6)$$

It has dimension 18 and contains no cross-products. Each coordinate is standardized over the 64-profile library before selection.

4.3 Replicated source scores

For source task q , profile j , and replication r , let Y_{qjr}^f and Y_{qjr}^g be the two simulator outputs. Define

$$\hat{f}_{qj} = R^{-1} \sum_{r=1}^R Y_{qjr}^f, \quad (7)$$

$$\hat{m}_{qj} = \bar{Y}_{qj}^g + z_{1-\alpha_q} \max\{s_{qj}^g, 10^{-3}\} - \tau_q, \quad (8)$$

where s_{qj}^g is the sample standard deviation with divisor $R - 1$. Within each task, deterministic average-tie percentile ranks map the 64 objective means and margins to $[0, 1]$, with lower values preferred. Let r_{qj}^f and r_{qj}^g denote those ranks. The primary domain-blind scores are

$$r_j^f = \frac{1}{m} \sum_{q=1}^m r_{qj}^f, \quad r_j^g = \frac{1}{m} \sum_{q=1}^m r_{qj}^g. \quad (9)$$

Whether $\hat{m}_{qj} \leq 0$ in every source task is logged as a diagnostic but does not filter the library. Descriptor-conditioned source weights are a secondary control and are not part of the primary method.

Table 1: Frozen primary method specification.

Component	Specification
Decision class	Ordered bounded profile $h : [0, 1] \rightarrow [0, 1]$ on a declared grid
Library	64 outcome-free profiles; 128 reference nodes; seed 20260808
Source archive	2 tasks $\times$ 64 profiles $\times$ 3 replications = 384 calls
Profile coordinate	Frequencies 0:8, penalty .25, diagonal squares, 18 dimensions
Source score	Equal-task average of objective and Gaussian chance-margin percentile ranks
Selection	First center $.5r^g + .5r^f$; nine Gonzalez farthest-first centers
Metric weights	Profile coordinates standardized; safety and objective rank weights both 1
Target map	Linear interpolation to declared nodes; deterministic integer rounding
Target information	Grid, bounds, dimension, and declared ordering; no outcomes or oracle quantities

4.4 Rank-seeded farthest-first selection

Let z_j be the standardized version of $\phi(h_j)$. This 18-dimensional coordinate contains structural profile information only. Selection uses the 20-dimensional augmented coordinate

$$\eta_j = (z_j, r_j^g, r_j^f) \quad (10)$$

with Euclidean distance. The first center is

$$j_1 \in \arg \min_j \{ .5r_j^g + .5r_j^f \}, \quad (11)$$

breaking ties by r_j^g , then r_j^f , then stable profile identifier. For $\ell = 2, \dots, n_0$,

$$j_\ell \in \arg \max_{j \notin \{j_1, \dots, j_{\ell-1}\}} \min_{s < \ell} \|\eta_j - \eta_{j_s}\|_2, \quad (12)$$

with the lowest library index breaking an exact tie. The primary design uses $n_0 = 10$. Each selected 128-node profile is linearly interpolated onto the declared target nodes, multiplied by L_q , clipped, and deterministically rounded to an integer policy.

The distinction between z and η is substantive. Source ranks affect which library members are selected, but the target transfer condition in Section 5 is stated only in structural z -space. The coordinate projection $\pi_z(\eta) = z$ is nonexpansive under the Euclidean norms, so any radius achieved in augmented space is also a valid, possibly loose, radius in structural space. No target analogue of a source percentile rank is assumed.

4.5 Search and verification interfaces

The initial design is backend independent. The primary attribution experiment sets $N = n_0 = 10$, so it contains no sequential optimizer and isolates the frontend. Functional SCBO, SAASBO, and native transfer methods are separate comparisons.

Given noisy target records, the neutral protocol freezes up to three distinct shortlist entries in this order: best observed-objective empirical feasible policy, observed safe-interior policy, and minimizer of

$$Y^f(x) + 2[Y^g(x) - \tau]_+, \quad (13)$$

followed by objective-ranked fallbacks if needed. Fresh simulator replications then test shortlist members sequentially. Candidate j with budget v_j is certified only if all v_j constraint replications satisfy the threshold and $(1 - \alpha)^{v_j} \leq \delta/K_s$. The primary budgets are $(80, 80, 80)$, $K_s = 3$, and $\delta = .05$. Failure to certify produces abstention, not a deployment.

4.6 Source-free counterfactuals

The most important control is *generic DCT maximin*. It uses the same 64 profiles, the same ϕ , the coordinate-space medoid as its first center, and farthest-first for the remaining nine, but no outcomes. Random low-frequency, natural three-block, and raw Sobol designs separate other forms of structural prior. The finite-library oracle is labeled as unattainable and appears only in audit artifacts.

For clarity, the method is a source-scored subset of a preconstructed library. It is not described as discovering the library from data, and its value over generic structure is an empirical question rather than a premise.

5 Finite-Sample Guarantees and Scope

The theory explains the finite implementation under explicit conditions. It does not assert that every target is aligned with the source population.

5.1 Profile-grid consistency

Let $c_k(h) = \int_0^1 h(t)b_k(t) dt$, where $b_0(t) = 1$ and $b_k(t) = \sqrt{2}\cos(\pi kt)$ for $k \geq 1$. Let $\hat{h}_d$ be a reconstruction from d ordered samples.

Proposition 5.1 (Coefficient stability). *If $\|h - \hat{h}_d\|_\infty \leq \epsilon_d$ and $\|b_k\|_\infty \leq B_k$, then*

$$|c_k(h) - c_k(\hat{h}_d)| \leq \epsilon_d B_k.$$

If h is L_h -Lipschitz and $\hat{h}_d$ is the regular-midpoint piecewise-constant reconstruction, then $\epsilon_d \leq L_h/(2d)$. Consequently the error of the frequency-penalized coefficient is at most

$$\frac{L_h B_k}{2d(1 + \kappa k)}.$$

Linear interpolation from a reference grid of size d_{ref} has the analogous bound $L_h/(2d_{\text{ref}})$ before deterministic rounding.

The proof applies the integral triangle inequality and the nearest-node or linear-interpolation error. It justifies comparison across discretizations of one regular profile; it does not turn an unordered vector into a profile.

5.2 Finite-library coverage

For any coordinate u on the finite library, define the covering radius of $A \subset \mathcal{L}$ by

$$r(A; u) = \max_{h \in \mathcal{L}} \min_{a \in A} \|u(h) - u(a)\|_2.$$

Theorem 5.2 (Farthest-first finite k -center bound). *Let A_k be the k centers produced by Gonzalez farthest-first from any declared first center, and let r_k^* be the smallest covering radius attainable by any k library members. Then*

$$r(A_k; u) \leq 2r_k^*.$$

The standard witness proof uses the k centers and one farthest unselected point. Two of these $k + 1$ pairwise-separated witnesses must share an optimal cluster. The implementation emits this witness certificate for every design. For the implemented $\eta = (z, r^g, r^f)$, Euclidean projection gives

$$r(A_k; z) \leq r(A_k; \eta) \leq 2r_k^*(\eta). \tag{14}$$

Thus outcome ranks may alter selection, while the resulting set still has a checkable structural-space cover. Equation (14) does not assert that source ranks themselves transfer to a target.

5.3 Source-score uncertainty

Let s_{qj} denote either a true source objective statistic or a true source chance-margin statistic and $\hat{s}_{qj}$ its replicated estimate.

Proposition 5.3 (Separated-rank recovery). *Suppose $|\widehat{s}_{qj} - s_{qj}| \leq r_q$ simultaneously over a finite source library. Every pair satisfying $|s_{qj} - s_{q\ell}| > 2r_q$ has the correct estimated order.*

For the Gaussian simulator used in the registered synthetic study, let $R \geq 2$, $\nu = R - 1$, and let $q_{\nu,p}$ be the p quantile of χ_ν^2 . For one profile with true mean μ and scale σ , define

$$r_\mu(\delta_\mu) = \sigma \sqrt{\frac{2 \log(2/\delta_\mu)}{R}}, \quad (15)$$

$$r_\sigma(\delta_\sigma) = \sigma \max \left\{ 1 - \sqrt{\frac{q_{\nu, \delta_\sigma/2}}{\nu}}, \sqrt{\frac{q_{\nu, 1-\delta_\sigma/2}}{\nu}} - 1 \right\}. \quad (16)$$

The normal mean tail and Cochran identity $(R-1)S^2/\sigma^2 \sim \chi_{R-1}^2$ imply, with probability at least $1 - \delta_\mu - \delta_\sigma$,

$$|\widehat{m} - m| \leq r_\mu(\delta_\mu) + z_{1-\alpha} r_\sigma(\delta_\sigma), \quad \sigma \geq 10^{-3}. \quad (17)$$

The variance floor is harmless here because $a \mapsto \max\{a, 10^{-3}\}$ is one-Lipschitz. Allocating the two tail probabilities across the finite set of tasks, profiles, and score heads gives the simultaneous radius required by Proposition 5.3. This result does not promise exact recovery of nearly tied ranks from three replications; the sensitivity matrix therefore varies replications and library size rather than treating source order as known. Lean checks the finite-sum variance identity, floor-aware error propagation, bad-event union, and rank implication. The normal and chi-square sampling laws are the standard external distributional identities used to instantiate those checked events (Cochran, 1934).

5.4 Conditional source-to-target coverage

Let $m_q(h)$ be the target chance margin, with $m_q(h) \leq 0$ implying feasibility. Let z_q^* be an ideal target structural coordinate and $\widehat{z}$ the implemented structural projection of η . The following statement exposes the assumption that source and target share useful profile geometry; it imposes no target interpretation on (r^g, r^f) .

Assumption 5.4 (Aligned profile geometry). There exist a target safe-center profile h_q^* , a library member h_s , and constants $\gamma_q, \Delta_q, \epsilon_\eta, L_q \geq 0$ such that

$$\begin{aligned} m_q(h_q^*) &\leq -\gamma_q, \\ \|z_q^*(h_s) - z_q^*(h_q^*)\|_2 &\leq \Delta_q, \\ \|\widehat{z}(h) - z_q^*(h)\|_2 &\leq \epsilon_\eta \quad \text{for the profiles used below,} \\ |m_q(h) - m_q(h')| &\leq L_q \|z_q^*(h) - z_q^*(h')\|_2. \end{aligned}$$

Theorem 5.5 (Aligned atlas coverage). *Let $\mathcal{A}_{n_0}$ cover $\mathcal{L}$ within radius $r_{n_0}^\eta$ in augmented η . By (14) it covers the same support within that radius in $\widehat{z}$. Under Assumption 5.4, if*

$$L_q(r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta) \leq \gamma_q, \quad (18)$$

then $\mathcal{A}_{n_0}$ contains a target-feasible policy.

The proof projects the augmented cover to $\widehat{z}$, chooses the atlas member covering h_s , transfers the distance to $z_q^\star$ by two coordinate-error terms, and applies Lipschitz margin and safe depth. Equation (18) is useful precisely because it can fail: frequency shifts, irregular grids, and misspecified targets raise Δ_q or ϵ_η , while shallow feasible basins reduce γ_q .

Figure 2 gives the geometric interpretation. The symbol ϵ_η denotes error between the implemented projected structural coordinate $\widehat{z}$ and the ideal target coordinate $z_q^\star$; it does not transfer the source-rank components (r^g, r^f) to the target. The circle labeled r_{n_0} depicts the structural-space upper bound obtained by projecting the augmented-coordinate cover $r_{n_0}^\eta$.

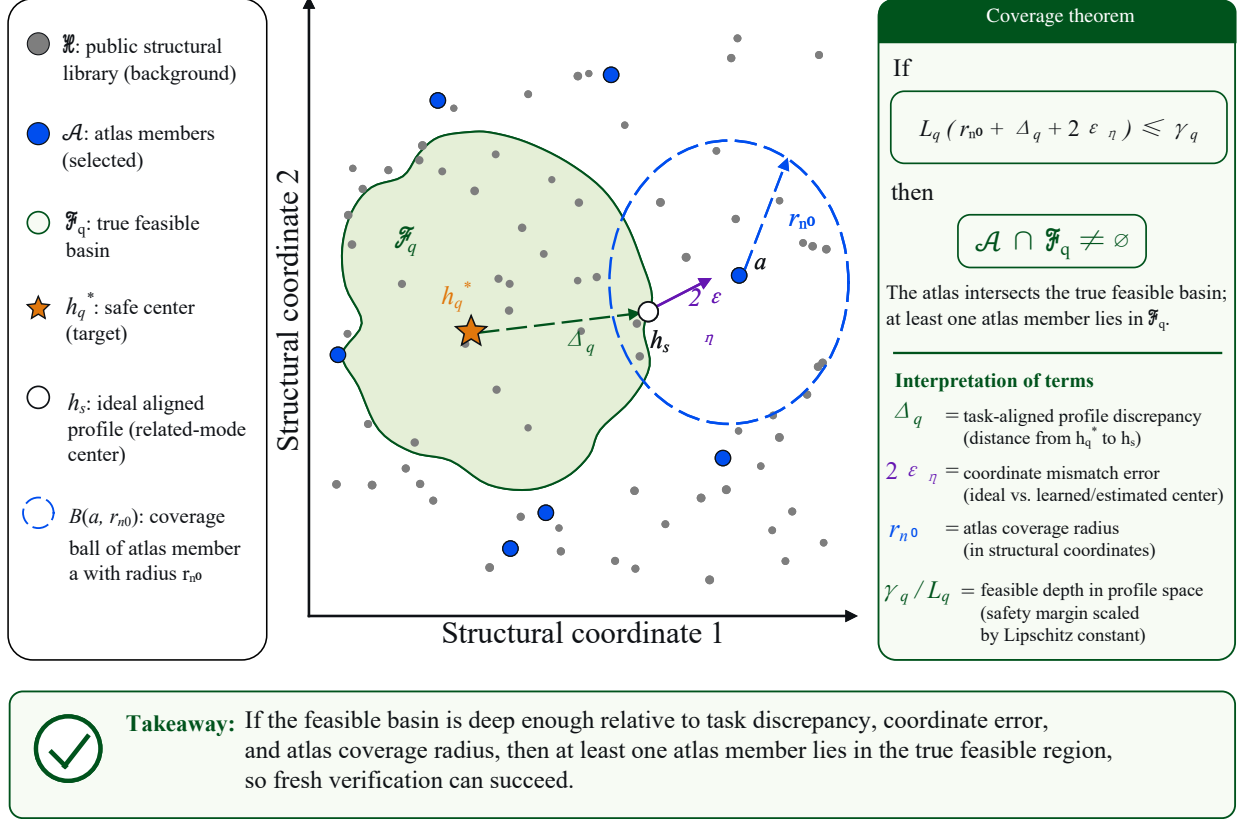


Figure 2: Schematic geometry of atlas coverage in structural profile space. Gray points represent the public profile library and blue points the selected atlas. The target has an unknown feasible basin $\mathcal{F}_q$ containing a safe center h_q^* . A source-supported library profile h_s lies within task discrepancy Δ_q of the safe center; projected coordinate mismatch contributes at most $2\epsilon_\eta$; and an atlas member a covers h_s within the projected upper bound r_{n0} . Hence $L_q(r_{n0} + \Delta_q + 2\epsilon_\eta) \leq \gamma_q$ implies $\mathcal{A} \cap \mathcal{F}_q \neq \emptyset$. The diagram illustrates the coverage theorem and is not an empirical two-dimensional projection of a target instance.

The confirmatory generator uses eight fixed, equally weighted regime strata, not one pooled IID law. At a fixed resolution d , let stratum s have n_s independent held-out task seeds, weight w_s with $\sum_s w_s = 1$, and hit indicators $H_{si} \in [0, 1]$. Weighted Hoeffding concentration, which does not require identical distributions across strata, gives with probability at least $1 - \delta$,

$$\sum_s w_s \mathbb{E}[H_s] \geq \sum_s \frac{w_s}{n_s} \sum_{i=1}^{n_s} H_{si} - \sqrt{\frac{\log(2/\delta)}{2} \sum_s \frac{w_s^2}{n_s}}. \quad (19)$$

Here $w_s = 1/8$ and $n_s = 20$. The deterministic component `task_seed mod 5` is an induced latent category whose realized counts need not be balanced; independence and boundedness, rather than identical distributions, are sufficient. The same 160 latent task seeds are crossed with all three resolutions, so (19) is applied separately at each d ; the pooled 480-cell rate is

descriptive. The probability refers only to a prospective repetition of the registered stratified generator and requires the atlas to be frozen before held-out tasks. It is not a guarantee for arbitrary future domains.

5.5 Exact terminal verification

Fix a shortlist before verification. Candidate j receives v_j fresh iid Bernoulli indicators $Z_{jr} = 1\{Y_{jr}^g \leq \tau\}$ and is certified only when $\sum_r Z_{jr} = v_j$ and

$$p_0^{v_j} \leq \delta_j, \quad p_0 = 1 - \alpha, \quad (20)$$

where $\delta_j = \delta/K_s$ in the primary protocol.

Theorem 5.6 (Candidate and familywise validity). *For any fixed unsafe candidate with true feasibility probability $p_j < p_0$,*

$$\mathbb{P}\{\text{candidate } j \text{ is certified}\} = p_j^{v_j} \leq p_0^{v_j} \leq \delta_j.$$

For a frozen shortlist, regardless of dependence among the candidate tests,

$$\mathbb{P}\{\text{an unsafe policy is deployed}\} \leq \sum_{j=1}^{K_s} \delta_j \leq \delta.$$

Sequentially stopping at the first certificate does not alter this union-bound argument because the shortlist and all budgets are fixed before testing.

For $p_0 = .95$, $K_s = 3$, and $\delta = .05$, the minimum all-success budget is 80 per candidate. Its power at true feasibility probability p is p^{80} ; this is only 0.017 at $p = .95$. Validity therefore does not imply nonvacuity, and Section 7 reports both. The exact Clopper–Pearson lower bound is used only for the preregistered sensitivity calculation (Clopper and Pearson, 1934).

5.6 Formalization boundary

The profile consistency, finite k -center, rank-recovery, aligned-coverage, task-law, binomial, and budget implications are formalized in Lean 4 without `sorry`, `admit`, or project-defined `axiom`. Machine checking establishes that the finite conclusions follow from their premises. It does not establish that Assumption 5.4 holds in Energy or any unspecified target population; those questions are empirical and the negative experiments are retained.

Table 2: Registered stress regimes. The target generator hides its basis, center, safe radius, and task seed from every algorithm.

Regime	Default rank	Principal departure
Aligned low frequency	8	Source and target share low-frequency support
Growing effective rank	24	More active coefficients at fixed target budget
Frequency support shift	8	Target frequencies shifted beyond source support
Coordinate permutation	8	Ordered values are permuted before implementation
Irregular grid	8	Nonuniform declared nodes
Piecewise smooth	12	Blockwise basis rather than cosine basis
Sparse high frequency	16	Few active high-frequency coordinates
Misspecified target	24	Target center independent of source centers

6 Experimental Design

The experiment program was frozen before confirmatory outcomes. A compact registry commits the protocol, method specification, result-cell hashes, failures, and analysis endpoints. All non-oracle methods use the same target truth only after deployment to compute audit metrics.

6.1 Randomized ordered-profile stress test

The primary study uses eight fixed, equally weighted regime strata. Within each stratum, independent hash-derived task-seed streams generate held-out latent tasks. Table 2 separates nominal grid resolution from latent complexity.

For each regime, we draw 20 independently seeded latent target tasks and cross the same seeds with $d \in \{200, 1000, 10000\}$. This gives 160 independent latent task seeds and 480 correlated task-resolution cells per design. Each cell is paired across six designs: source-scored, generic DCT maximin, random low frequency, natural three-block, raw Sobol, and an explicitly labeled finite-library oracle. The primary non-oracle matrix has $8 \times 3 \times 20 \times 5 = 2400$ cells; including the oracle gives 2880. Every design uses $n_0 = N = 10$, so the primary comparison is a pure initial-design intervention. The source-scored arm pays 384 source calls; source-free arms pay none. All use the same frozen-shortlist verifier.

For each synthetic target regime, the two supplied source tasks are generated from that same regime with the role set to source. This is a regime-matched transfer experiment, not a source-retrieval experiment. Algorithms receive neither target outcomes nor hidden target geometry, but the experimenter has already supplied a related archive. The Energy region holdout is the explicit archive-mismatch control.

The primary chance level is $\alpha = .05$, and task safe mass is calibrated to .08 on a fixed outcome-free audit library distinct from the design library. Regret is measured against the best feasible policy in that audit library. The $d = 10000$ cells refine the grid while holding the registered latent law fixed unless the regime explicitly changes rank.

6.2 Sensitivity and information controls

An 8640-cell one-factor-at-a-time matrix varies active rank, α , safe mass, n_0 , number of source tasks, profiles per task, source replications, retained frequency, frequency penalty, source-rank metric weights, and the first-center safety weight. These are diagnostics, not a post-confirmatory hyperparameter search. A separate 1280-cell matrix crosses declared versus schema-blind profile mapping with equal versus descriptor-conditioned source weights. The latter weighting is excluded from the primary method regardless of its result.

6.3 Cost and optimizer controls

The equal-preverification-cost matrix compares 384 source plus 10 target calls with 394 target-only calls for each source-free design. The same independent verifier is applied afterward and all-in realized calls include its cost.

Target-only functional SCBO searches an 8-coefficient cosine space with the published constrained trust-region posterior-sampling backend (Balandat et al., 2020; Eriksson and Poloczek, 2021). We test coefficient counts 4, 8, and 16 at $N = 13$, and the primary 8-coefficient version at $N = 394$. One $N = 394$ task in the irregular-grid regime returned duplicate canonical posterior samples; it remains a failed, unsuccessful cell.

Eight native transfer pipelines—Safe F-PACOH, RGPE, stacked transfer GP, multi-task GP, FSBO, HyperBO, MetaBO, and MALIBO—receive the same 384-call source archive, $n_0 = 10$, $N = 20$, and terminal verifier, but retain their own prior, representation, initialization, and acquisition logic. This is an end-to-end transfer comparison, not a common-atlas backend comparison.

6.4 Electricity-reserve negative external control

The external study uses the Open Power System Data time-series release (Open Power System Data, 2020). A policy maps 1000 ordered forecast-stress quantiles to reserve responses over a 168-hour operational horizon. Energy V3 includes 18 markets in five geographic regions and five algorithmic seeds per held-out market. All source markets from the target’s region are excluded. The source arm uses 768 source calls, $N = 13$ target calls, and the same

public profile library; generic DCT, random low frequency, natural constant, raw Sobol, and target-only functional SCBO are controls.

Fresh verification windows are sampled from a held-out empirical calendar split. The exact-binomial claim is therefore conditional on the declared replication distribution, not a theorem about an unobserved future year. Postdecision audits test chronological blocks and physically nonoverlapping windows, but are descriptive because markets share calendar data and only five regions exist. Energy V2 is retained as a separate historical specification; V3 is the final external control. No Energy result was used to modify the frozen atlas.

6.5 Inference and failure accounting

For randomized profiles, a latent task seed within a fixed regime and resolution is the generalization unit; paired bootstrap intervals resample these seeds. Task-law concentration is applied separately at each resolution with regimes as fixed strata. Rates pooled over all three resolutions describe 480 task–resolution cells but are not treated as 480 independent tasks. Simulation seeds within a fixed external market measure algorithmic repeatability, while geographic regions are the conservative Energy comparison units. We do not pool seeds as if they were independent domains. Exact sign tests are Holm-adjusted within their registered families, but effect sizes and intervals receive priority over very small p -values. Every timeout, duplicate-sampling failure, and false certificate remains in its denominator.

Primary endpoints are feasible coverage, certified true-feasible deployment, false certification, and penalized loss. We additionally report source, search, verification, unamortized all-in, and $M = 20$ amortized calls; conditional regret is secondary.

7 Results

7.1 Aggregate randomized-task performance

Table 3 and Figure 3 report the primary randomized task law. The source-scored design contains a truly feasible policy in 91.7% of tasks and yields a certified true-feasible deployment in 47.3%. Generic DCT maximin reaches 37.5% and 10.6%; raw Sobol reaches 80.2% and 30.0%. Penalized loss is 0.163 for source scoring, 0.746 for generic DCT, and 0.337 for Sobol. Thus source outcomes contribute beyond the shared structural library on this registered mixture.

The paired source-minus-generic-DCT certified-success differences are 0.406 (bootstrap 95% CI [.325, .488]) at $d = 200$, 0.356 ([.275, .438]) at $d = 1000$, and 0.338 ([.256, .419]) at $d = 10000$. These stable differences support cross-grid transfer for the registered profile

Table 3: Primary randomized profile results. Each row contains 480 paired task–resolution cells: 160 independently seeded latent tasks crossed with three grid resolutions.

Initial design	Tasks	Feasible (%)	Certified (%)	False cert.	Penalized loss
Generic DCT maximin	480	37.5	10.6	3	0.746
Natural blockwise	480	83.8	19.4	0	0.278
Random low frequency	480	50.8	13.5	0	0.610
Raw Sobol	480	80.2	30.0	2	0.337
Source-scored atlas	480	91.7	47.3	0	0.163

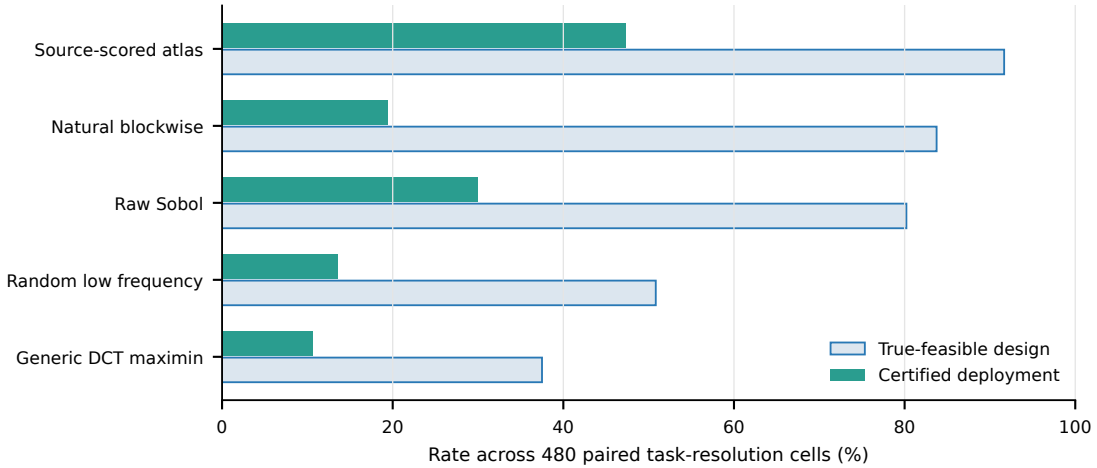


Figure 3: Aggregate feasible coverage and independently certified deployment. Bars aggregate task–resolution cells; inferential comparisons are paired within resolution and do not treat the three grids as independent tasks.

law. They do not establish increasing difficulty with d : the $d = 10000$ experiment is a discretization-refinement test.

7.2 Where source scoring fails

The aggregate advantage is not uniform. Table 4 and Figure 4 preserve every adverse regime. Raw Sobol outperforms source scoring in aligned low frequency and irregular grids; natural structure is slightly better under sparse high-frequency activity; and neither method certifies useful policies under full misspecification. Source scoring is strongest for permutation, piecewise smoothness, growing rank, and the registered frequency shift. In the sparse-high-frequency regime its feasible coverage falls from 20/20 tasks at $d = 200$ to 0/20 at $d = 10000$, despite unchanged target budget. This is direct evidence against a generic high-dimensional claim.

The source arm records no false certificates in 480 task–resolution cells. Generic DCT

Table 4: Rates aggregated over three grid resolutions within each regime. The comparator is the best source-free design in that row.

Regime	Source feasible	Source certified	Best source-free certified	Difference
Aligned low frequency	100.0	56.7	Raw Sobol: 91.7	-35.0
Growing rank	100.0	40.0	Random low frequency: 20.0	+20.0
Frequency shift	95.0	60.0	Natural blockwise: 25.0	+35.0
Permutation	100.0	90.0	Random low frequency: 50.0	+40.0
Irregular grid	100.0	15.0	Raw Sobol: 38.3	-23.3
Piecewise smooth	100.0	95.0	Raw Sobol: 50.0	+45.0
Sparse high frequency	63.3	21.7	Natural blockwise: 25.0	-3.3
Misspecified target	75.0	0.0	Raw Sobol: 1.7	-1.7

Table 5: Information-contract controls at $d = 1000$.

Schema	Source weighting	Tasks	Feasible (%)	Certified (%)	Loss
declared	domain blind	160	95.0	46.2	0.143
declared	conditioned	160	85.0	44.4	0.258
schema blind	domain blind	160	92.5	35.0	0.180
schema blind	conditioned	160	82.5	33.1	0.296

and raw Sobol record three and two, respectively. Target-only functional SCBO at $N = 13$ certifies 15 of 480 primary cells with three false certificates, all in the frequency-support-shift regime. Increasing its cosine rank does not remove the general coverage deficit.

The aligned-low-frequency reversal has a concrete geometric explanation. Figure 5 reconstructs each frozen design and accesses the hidden target safe center only afterward. Across the 600 design points per arm, raw Sobol has median latent-center distance 0.0343 and median non-DC low-frequency energy 0.0258, compared with 0.0813 and 0.2292 for source scoring. Its pointwise true-feasible rate is 94.7%, versus 10.0% for both source and generic DCT. High-dimensional independent coordinates average toward a near-constant profile with DC coefficient .5; the aligned generator’s safe center happens to occupy that region. The diagnostic explains the registered reversal but is postdecision, uses oracle geometry, and neither changes outcomes nor supports method selection.

7.3 Schema and sensitivity

Declared profile ordering is valuable information. With domain-blind source weights, removing the declared schema reduces certified success from 46.2% to 35.0% (Table 5). Descriptor-conditioned weighting reduces both declared- and blind-schema performance and is not promoted. Source scoring therefore does not need the target family label, but it does rely on an interpretable ordered profile map.

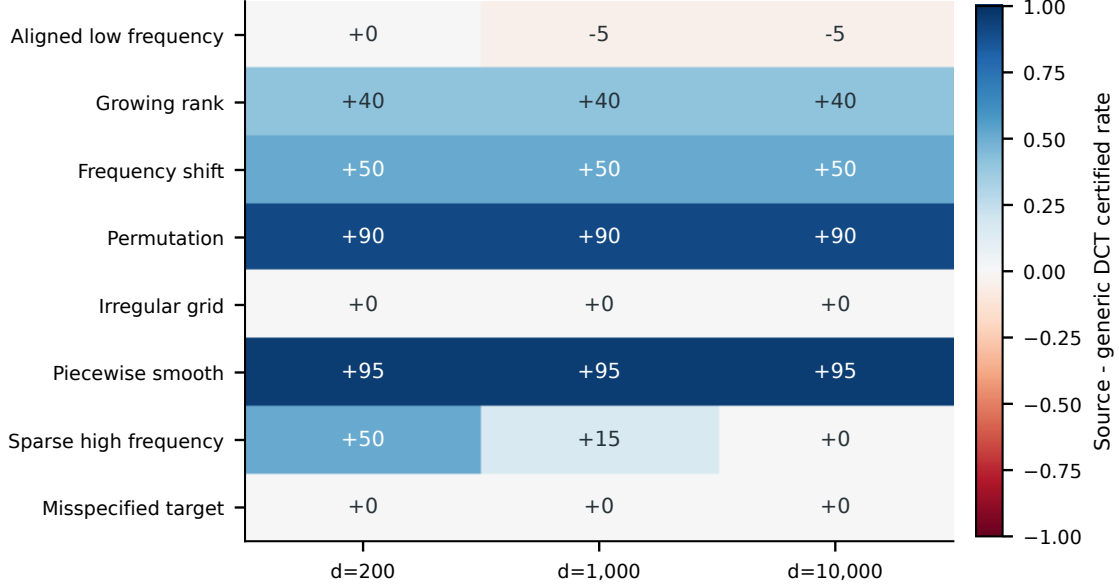


Figure 4: Source-minus-generic-DCT certified-success differences by regime and grid resolution. Positive aggregate performance coexists with registered reversals.

Sensitivity is nonmonotone, which argues against treating source volume as a free advantage. Raising profiles per source task from 64 to 128 reduces certified success from 46.2% to 38.8%; using one, three, or five replications gives 45.0%, 46.2%, and 45.6%. Four source tasks give 45.0%, not more than two. Retaining 16 frequencies reaches 48.1%, while four reaches 43.1%. Increasing n_0 from 10 to 20 yields only 46.9% certification, while $n_0 = 5$ yields 42.5%. The first-center safety weight is more consequential: .25 reduces certification to 32.5%, whereas .5 and .75 give 46.2% and 45.0%. Figure 6 summarizes the source-budget axes.

7.4 Cost changes the ranking

At the source method’s primary budget, mean unamortized all-in calls are about 559 across the randomized law, while amortization over 20 targets reduces the source contribution from 384 to 19.2 calls per target. Comparable primary source-free designs use roughly 213–233 calls because verification dominates their ten search calls. Figure 7 makes both accounting views explicit.

When the 384 source calls are instead given to target-only search, the source method is not best (Table 6). It certifies 46.2% of 160 tasks. Generic DCT and raw Sobol each certify 53.8%, natural structure 53.1%, and target-only functional SCBO 58.1%. The latter includes one failed cell as an unsuccessful outcome. Mean realized all-in costs are already close because methods consume different verification effort. The source archive is thus an

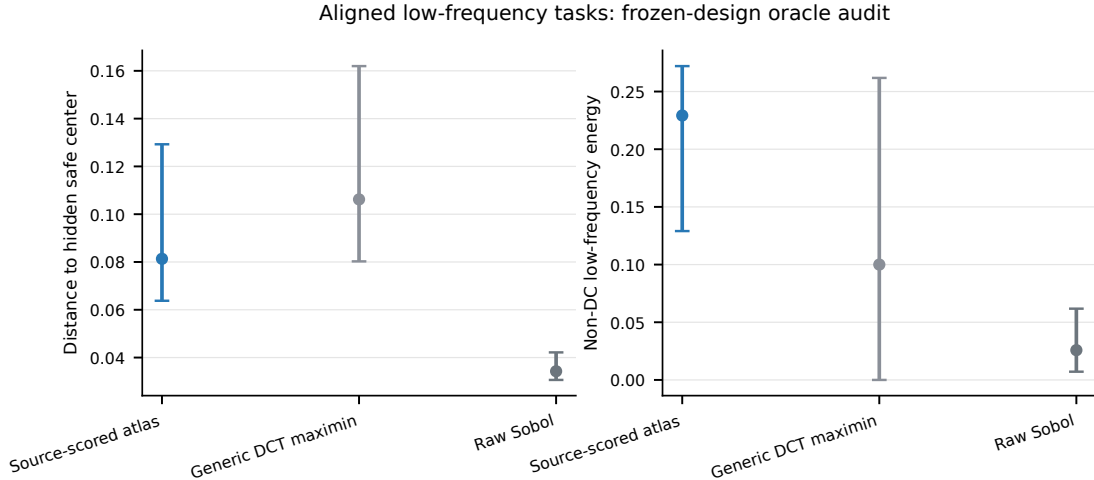


Figure 5: Postdecision aligned-regime oracle audit. Points show medians and bars show 10th–90th percentiles over frozen design points. The hidden safe center was unavailable to every algorithm.

Table 6: Equal preverification cost at $d = 1000$: 384 source plus 10 target calls versus 394 target-only calls. Verification remains additional and is included in the final column.

Method	Source	Target	Certified (%)	False	Algo. fail.	Mean all-in
Generic DCT maximin	0	394	53.8	0	0	564.5
Natural blockwise	0	394	53.1	0	0	564.5
Random low frequency	0	394	45.6	0	0	584.0
Raw Sobol	0	394	53.8	0	0	564.0
Source-scored atlas	384	10	46.2	0	0	562.0
Target-only functional SCBO	0	394	58.1	0	1	549.1

amortized multi-target investment, not a one-target total-cost win.

Using primary mean call counts, the source archive breaks even in simulator calls after roughly 7 target deployments versus generic DCT, 9 versus natural structure, and 11 versus raw Sobol. These are cost thresholds, not guarantees of outcome dominance; Energy demonstrates why the distinction matters.

Table 7 also divides amortized calls by the observed probability of a true-feasible certificate. In the primary matrix, source scoring uses 411.3 expected calls per certified deployment when the archive is shared by 20 targets, versus 2192.9 for generic DCT, 1143.2 for natural structure, and 709.4 for raw Sobol. In the equal-preverification matrix the corresponding values are 426.4, 1050.2, 1062.6, and 1049.3. This descriptive metric favors source reuse after roughly two targets in the latter comparison even though source scoring loses the single-target equal-cost success rate. It conditions the economic calculation on certification success; it does not price unsafe deployment, abstention, or objective quality.

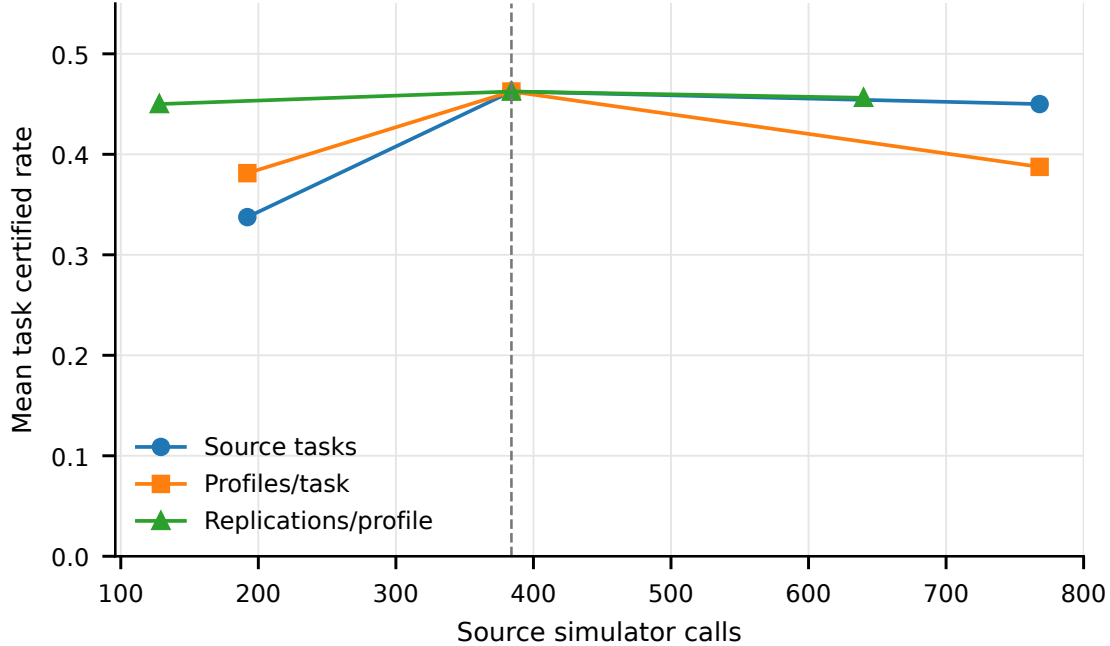


Figure 6: Source-budget sensitivity. More tasks, profiles, or replications do not monotonically improve the frozen design.

7.5 External Energy result

Energy is a negative external control for source learning. Table 8 shows that source scoring certifies 60/90 runs, while generic DCT maximin certifies 70/90. Random low frequency, natural constants, raw Sobol, and functional SCBO certify 61, 57, 52, and 58. No arm records a false certificate. Across five regions, the source-minus-generic safe-rate difference is -0.079 with bootstrap 95% CI $[-.137, -.020]$: zero region wins, three losses, and two ties. Source scoring is not rescued by a favorable objective comparison; median objective is 0.0345 for source scoring and 0.0455 for generic DCT among certified summaries, but the sets of certified runs differ and conditional medians cannot override the coverage reversal.

The temporal audit preserves 312 of 358 V3 certificates under both block and nonoverlap conditions. It is descriptive, not an iid future-calendar certificate. The external conclusion is consequently narrow but useful: ordered low-frequency structure helps relative to raw Sobol, while source outcomes can mis-rank that structure under geographic and temporal shift.

7.6 Verifier power and native transfer

The verifier is valid but intentionally conservative. At 80 all-success replications, certification power is only 1.7% at the threshold $p = .95$, 13.2% at .975, and 44.8% at .99 (Figure

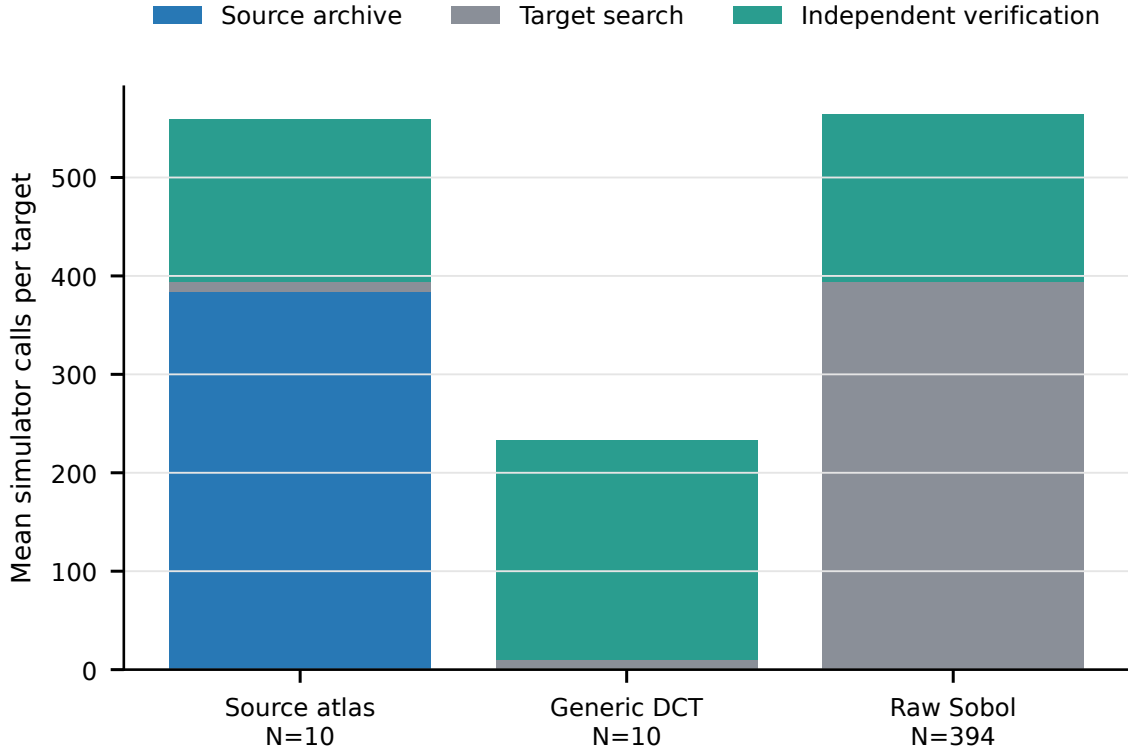


Figure 7: Mean source, target-search, and verification calls under unamortized and 20-target accounting.

8). Therefore the gap between 91.7% feasible coverage and 47.3% certified deployment is expected; an empty certificate set is not evidence that search found no feasible point. A Clopper–Pearson threshold rule is reported as sensitivity, not substituted after observing outcomes.

In the native transfer matrix, RGPE certifies 30/60 tasks and MetaBO 28/60; the other six methods range from 0/60 to 14/60 (Online Supplement, Table S4). All eight certify 0/20 Queue tasks. These results show that ordinary end-to-end transfer remains difficult, but they do not establish a broad SOTA ranking from three historical synthetic families.

Finally, cumulative factor-HVD substantially improves held-out log-variance RMSE and variance-shape correlation relative to pooled variance, yet does not improve feasible recovery, regret, false certification, or verification cost under matched designs. It is therefore retained only as a calibration diagnostic in the Online Supplement, Section C, not as a main optimization contribution.

Table 7: Outcome-adjusted simulator cost. Calls per success equal mean amortized all-in calls divided by the observed true-feasible certification rate. Break-even compares source scoring with each source-free row.

Matrix	Design	Certified (%)	Calls/success ($M = 1$)	Calls/success ($M = 20$)	Break-even M
$N = 10$ primary	Generic DCT maximin	10.6	2192.9	2192.9	1
$N = 10$ primary	Natural blockwise	19.4	1143.2	1143.2	2
$N = 10$ primary	Random low frequency	13.5	1716.9	1716.9	1
$N = 10$ primary	Raw Sobol	30.0	709.4	709.4	3
$N = 10$ primary	Source-scored atlas	47.3	1182.7	411.3	—
Equal preverification	Generic DCT maximin	53.8	1050.2	1050.2	2
Equal preverification	Natural blockwise	53.1	1062.6	1062.6	2
Equal preverification	Random low frequency	45.6	1280.0	1280.0	1
Equal preverification	Raw Sobol	53.8	1049.3	1049.3	2
Equal preverification	Source-scored atlas	46.2	1215.1	426.4	—

Table 8: Energy V3 region-held-out results over 18 markets and five seeds per market. Source scoring does not beat the generic structured control.

Design/backend	Certified	False	Median objective	Mean verify
Source-scored atlas	60/90	0	0.0345	145.8
Generic DCT maximin	70/90	0	0.0455	153.8
Random low frequency	61/90	0	0.0394	157.3
Natural constant grid	57/90	0	0.0391	174.2
Raw Sobol	52/90	0	0.0722	162.7
Target-only functional SCBO	58/90	0	0.0447	162.7

8 Discussion

8.1 What the evidence supports

The study supports a conditional proposition: when target feasibility is organized by profile geometry represented in the source population, source outcomes can improve a very small structural initial design. The randomized task law tests more than the three mechanisms used during method development, and its permutation, rank, piecewise, frequency-shift, and misspecification regimes make the condition falsifiable. Source scoring improves aggregate coverage, yet the regime table and Energy study identify concrete reversals.

This interpretation differs from claiming a generic high-dimensional optimizer. The raw vector may contain 10^4 coordinates, but the decision is an ordered profile and the library has finite structural complexity. Profile grid refinement is operationally relevant—a controller may require a fine lookup table—but it is not equivalent to adding 10^4 unrelated decision variables. The proper benefit is invariance to representation resolution under shared profile structure.

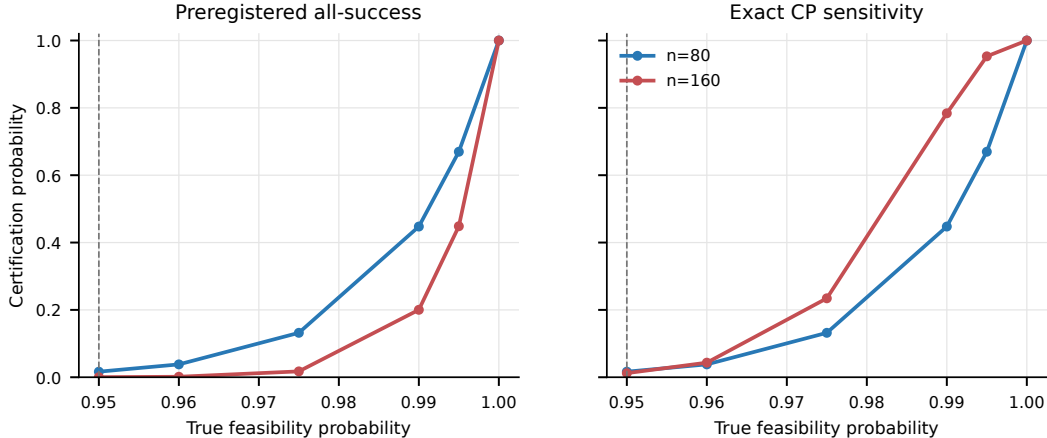


Figure 8: Exact fixed-budget certification power. The preregistered all-success rule is familywise valid but weak near the chance threshold.

8.2 Why Energy is necessary

Removing Energy would make the evidence cleaner and the scientific claim weaker. The external reversal shows that a generic structural design can be better than source scoring when region-held-out source rankings do not transfer. Because V3 was frozen before its outcomes, tuning the source score after this result would convert the holdout into development data. We therefore retain Energy as a negative control and interpret it as an applicability boundary.

The practical implication is a diagnostic choice. If historical tasks and a new target share an ordered policy semantics and source rank stability can be validated on genuinely held-out tasks, source scoring is reasonable. If the target is temporally or geographically shifted and generic low-frequency structure already has a credible physical basis, the generic maximin design is the safer default. The method should abstain from a transfer claim when these conditions cannot be assessed.

The current method begins after this assessment: a source archive is supplied exogenously, and the algorithm ranks profiles inside it. It does not retrieve related tasks from an unrestricted historical database. Regime-matched synthetic sources therefore identify the value of scoring a relevant archive, whereas Energy shows the cost of archive mismatch. Source retrieval and its own data-acquisition cost remain separate research and operational decisions.

8.3 Source cost as a portfolio investment

The equal-preverification-cost experiment prevents an appealing but incorrect story: 10 target calls do not beat 394 target calls for a single target. A source archive becomes economically

plausible only when reused. Let $C_0 = N_0 + V_0$ be the expected target-only calls per deployment and $C_A = N_A + V_A$ the source-design calls excluding the archive. The call-count break-even number of deployments is

$$M_{\text{break}} = \left\lceil \frac{S}{C_0 - C_A} \right\rceil, \quad C_0 > C_A. \quad (21)$$

In the randomized primary experiment, mean-call estimates give thresholds of about 7, 9, and 11 targets relative to generic DCT, natural structure, and raw Sobol. This arithmetic does not include differences in simulator setup, wall-clock parallelism, or policy quality, and Energy shows that paying the archive can still be inferior. It is a planning threshold, not a dominance theorem.

If certified deployment rather than a target instance is the unit of value, let p_A and p_0 be true-feasible certification probabilities. The corresponding descriptive efficiencies are

$$\tilde{C}_A(M) = \frac{S/M + C_A}{p_A}, \quad \tilde{C}_0 = \frac{C_0}{p_0},$$

and source scoring crosses a control when

$$M \geq \left\lceil \frac{S}{p_A C_0 / p_0 - C_A} \right\rceil, \quad p_A C_0 / p_0 > C_A.$$

This remains incomplete without decision-maker costs. A general operational loss would add c_{unsafe} per false deployment, c_{abstain} per abstention, and c_{quality} times objective regret to simulator calls. We report the components rather than choose those costs on the reader's behalf.

8.4 Verification is a separate operational decision

Independent verification frequently costs an order of magnitude more than target search. That is not overhead hidden by the method; it is the cost of a different claim. Search asks whether a promising policy has been found. Verification asks whether a frozen policy can be deployed with a specified false-certification bound. A target-only optimizer making the same safety claim must pay a comparable fresh-sample cost.

The all-success protocol is particularly conservative near the threshold. Its low power explains why feasible coverage greatly exceeds certified deployment. Increasing an all-success budget can even reduce power for any $p < 1$. Future deployments should preregister a more efficient exact Clopper–Pearson or sequential ranking-and-selection design rather than change the rule after inspecting results. Temporal applications also require block or mixing-aware

certification if iid replications are not physically defensible.

8.5 Benchmark fit and remaining threats

The frozen randomized generator reduces development overfitting but cannot erase it. The task law, profile library, and stress axes were authored within the project. The method also assumes declared ordered semantics, and the schema-blind drop confirms that this is a strong prior. Energy is genuinely different and unfavorable, but five geographic regions are too few for a broad population claim. The native transfer audit has only three historical families and all methods fail on Queue.

Other limitations are equally important. Related source tasks are assumed to have been supplied; source retrieval is not solved. Three source replications cannot reliably order near-tied profiles; source volume is nonmonotone; the finite library can miss a feasible basin entirely; and source rank aggregation can hide task disagreement. Objective comparisons among certified subsets are selection dependent. The randomized oracle measures regret relative to a finite audit library, not the unknown continuous optimum. One canonical functional-SCBO task failed and remains in the denominator. Finally, the formal theory proves finite implications conditional on alignment, within-stratum independence, and the declared replication law; it does not prove those assumptions from data. The same latent target seeds are reused across the three profile resolutions, so pooled 480-cell summaries are descriptive rather than a larger independent task sample.

These boundaries argue for a restrained deployment workflow: preregister the library and task law, keep a source-free structured control, validate rank stability on independent tasks, price source and verification calls, and retain abstention. Under those conditions, historical simulation is a transparent experimental-design asset rather than a hidden oracle.

9 Conclusion

We developed a source-scored structural initial design for chance-constrained ordered policy profiles. Replicated source outcomes rank a public library; farthest-first coverage selects ten profiles in a cross-grid coordinate; a replaceable backend searches; and an independent verifier certifies or abstains. Finite guarantees cover discretization, selection, rank recovery, conditional transfer, and verification.

On the registered task law, source scoring improves coverage and certification over generic structure, but reverses in adverse regimes and held-out Energy and loses at equal one-target cost. It is therefore an amortized, task-law-dependent investment, not a universal remedy.

References

- Satyajith Amaran, Nikolaos V. Sahinidis, Bikram Sharda, and Scott J. Bury. Simulation optimization: A review of algorithms and applications. *Annals of Operations Research*, 240(1):351–380, 2016. doi: 10.1007/s10479-015-2019-x.
- Bruce Ankenman, Barry L. Nelson, and Jeremy Staum. Stochastic kriging for simulation metamodeling. *Operations Research*, 58(2):371–382, 2010. doi: 10.1287/opre.1090.0754.
- Maximilian Balandat, Brian Karrer, Daniel R. Jiang, Samuel Daulton, Benjamin Letham, Andrew Gordon Wilson, and Eytan Bakshy. BoTorch: A framework for efficient Monte-Carlo Bayesian optimization. In *Advances in Neural Information Processing Systems*, volume 33, pages 21524–21538, 2020.
- Justin Boesel, Barry L. Nelson, and Seong-Hee Kim. Using ranking and selection to clean up after simulation optimization. *Operations Research*, 51(5):814–825, 2003. doi: 10.1287/opre.51.5.814.16751.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- William G. Cochran. The distribution of quadratic forms in a normal system, with applications to the analysis of covariance. *Mathematical Proceedings of the Cambridge Philosophical Society*, 30(2):178–191, 1934. doi: 10.1017/S0305004100016595.
- David Eriksson and Martin Jankowiak. High-dimensional Bayesian optimization with sparse axis-aligned subspaces. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 493–503, 2021.
- David Eriksson and Matthias Poloczek. Scalable constrained Bayesian optimization. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pages 730–738, 2021.
- David Eriksson, Michael Pearce, Jacob Gardner, Ryan D. Turner, and Matthias Poloczek. Scalable global optimization via local Bayesian optimization. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Matthias Feurer, Benjamin Letham, and Eytan Bakshy. Scalable meta-learning for Bayesian optimization. In *ICML 2018 Workshop on Automated Machine Learning*, 2018.

- Michael C. Fu. Optimization for simulation: Theory vs. practice. *INFORMS Journal on Computing*, 14(3):192–215, 2002. doi: 10.1287/ijoc.14.3.192.113.
- Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985. doi: 10.1016/0304-3975(85)90224-5.
- L. Jeff Hong, Jun Luo, and Barry L. Nelson. Chance constrained selection of the best. *INFORMS Journal on Computing*, 27(2):317–334, 2015. doi: 10.1287/ijoc.2014.0628.
- Mark E. Johnson, Leslie M. Moore, and Donald Ylvisaker. Minimax and maximin distance designs. *Journal of Statistical Planning and Inference*, 26(2):131–148, 1990. doi: 10.1016/0378-3758(90)90122-B.
- Benjamin Letham, Brian Karrer, Guilherme Ottoni, and Eytan Bakshy. Constrained Bayesian optimization with noisy experiments. *Bayesian Analysis*, 14(2):495–519, 2019. doi: 10.1214/18-BA1110.
- Max D. Morris and Toby J. Mitchell. Exploratory designs for computational experiments. *Journal of Statistical Planning and Inference*, 43(3):381–402, 1995. doi: 10.1016/0378-3758(94)00035-T.
- Open Power System Data. Time series. Data package, version 2020-10-06, 2020. URL https://data.open-power-system-data.org/time_series/.
- Vera Pan, Alp Kucukelbir, Mitra Baratchi, and Jan van Gemert. MALIBO: Meta-learning for likelihood-free Bayesian optimization. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 39102–39134, 2024.
- Valerio Perrone, Huibin Shen, Matthias Seeger, Cedric Archambeau, and Rodolphe Jenatton. Learning search spaces for bayesian optimization: Another view of hyperparameter transfer learning. In *Advances in Neural Information Processing Systems*, volume 32, pages 12751–12761, 2019.
- Matthias Poloczek, Jialei Wang, and Peter I. Frazier. Warm starting Bayesian optimization. In *2016 Winter Simulation Conference*, pages 770–781, 2016. doi: 10.1109/WSC.2016.7822140.
- James O. Ramsay and Bernard W. Silverman. *Functional Data Analysis*. Springer, New York, 2 edition, 2005. doi: 10.1007/b98888.

- Jonas Rothfuss, Vincent Fortuin, Martin Josifoski, and Andreas Krause. PACOH: Bayes-optimal meta-learning with PAC-guarantees. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9116–9126, 2021.
- Jonas Rothfuss, Bhavya Sukhija, Tim Birchler, Parnian Kassraie, and Andreas Krause. Meta-learning reliable priors in the function space. In *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pages 237–265, 2023.
- Petru Tighineanu, Kathrin Skubch, Paul Baireuther, Attila Reiss, Felix Berkenkamp, and Julia Vinogradska. Transfer learning with Gaussian processes for Bayesian optimization. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 6152–6181, 2022.
- Ngo Anh Vien, Heiko Zimmermann, and Marc Toussaint. Bayesian functional optimization. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, pages 4171–4178, 2018. doi: 10.1609/aaai.v32i1.11830.
- Michael Volpp, Lukas P. Fröhlich, Kirsten Fischer, Andreas Doerr, Stefan Falkner, Frank Hutter, and Christian Daniel. Meta-learning acquisition functions for transfer learning in Bayesian optimization. In *International Conference on Learning Representations*, 2020.
- Zi Wang, George E. Dahl, Kevin Swersky, Chansoo Lee, Zachary Nado, Justin Gilmer, Jasper Snoek, and Zoubin Ghahramani. Pre-trained Gaussian processes for Bayesian optimization. *Journal of Machine Learning Research*, 25(212):1–83, 2024.
- Martin Wistuba and Josif Grabocka. Few-shot Bayesian optimization with deep kernel surrogates. In *International Conference on Learning Representations*, 2021.
- David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997. doi: 10.1109/4235.585893.